

Microbiota modulation and anti-obesity effects of fermented *Pyrus ussuriensis* Maxim extract against high-fat diet-induced obesity in rats

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ABSTRACT

Pyrus ussuriensis Maxim (Korean pear) has been used for hundreds of years as a traditional herbal medicine due to its strong phytochemical profile and pharmacological efficacy. In this study, we evaluated the anti-obesity potential of *Pyrus ussuriensis* Maxim extracts (PUE) and investigated the underlying mechanisms using a combination of *in vitro*, *in vivo*, and microbiota regulation approaches. In an adipogenesis assay, the fermented (F)PUE and non-fermented (NF)PUE significantly reduced the differentiation of 3T3-L1 preadipocyte in a dose-dependent manner with an IC₅₀ of 85.33 and 96.67 µg/mL, respectively. In a high-fat diet (HFD)-induced obese rat model (n = 8 animals/group), oral administration of FPUE additionally reduced the total body weight gain significantly. No difference in food intake was observed, however, between the control-chow diet, FPUE, and NFPUE-treated HFD rats. Adipose tissue mass and systemic insulin resistance were markedly reduced in FPUE-treated HFD rats, in a dose-dependent manner. Treatment with FPUE also greatly improved obesity-related biomarkers, including total cholesterol, leptin, active ghrelin, Total GIP, adiponectin, and proinflammatory cytokines. Moreover, FPUE significantly suppressed HFD-induced adipogenic genes expression, while increasing fatty acid oxidation-related genes expression. Additionally, FPUE treatment attenuated the HFD-induced *Firmicutes* proportion within the intestinal microbiota by regulating key metabolic pathways, thus enhancing microbial population diversity (e.g., increasing *Bacteroides*, *Bifidobacterium*, *Prevotella*, *Eubacterium*, and *Clostridium*). Together, these results reveal a strong anti-obesity potential of FPUE through adipogenesis, lipid metabolism, weight reduction, and microbiota regulation, raising the possibility of developing FPUE as a novel therapeutic agent to control obesity and obesity-associated metabolic disorders.

1. Introduction

Obesity is considered a low-grade chronic inflammation and an imbalance between energy intake and its expenditure that increases lipid storage in adipose tissues, as well as induces excessive triglyceride accumulation and adipocyte differentiation [1]. The high-fat diet (HFD)-induced obese rat model is widely studied considering that HFD is a major cause of obesity, while long-term HFD intake significantly

increases body weight [2]. In addition, HFD-induced obese rats were shown to develop more extensive adipose tissue and abdominal fat weight, but also to acquire insulin resistance and hyperleptinemia, typically associated with obesity and serious health conditions such as heart disease (hypertension), arteriosclerosis, diabetes, polycystic ovarian syndrome, respiratory complications, osteoarthritis and certain forms of cancer [3]. Consequently, obesity is considered to impair the quality of life and decrease the life span. Reducing the obesity rate is

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thus expected to not only reduce the risk of morbidity and mortality but also decrease obesity-associated diseases. The obesity rate is increasing worldwide, including in South Korea, stimulating research on how to reduce obesity in the population [4,5]. The prevalence of obesity among the Korean population has risen from 29.7 % in 2009 to 35.7 % in 2018, with the total cost of obesity estimated to reach 1.8 billion dollars based on the Korea National Health and Nutrition Examination Survey (KNHANES) [6].

A network of transcriptional factors that coordinate the expression of genes and morphological changes of cells regulates the lipogenesis-lipolysis balance. These include the master regulator of lipolytic genes, peroxisome proliferator-activated receptor γ (PPAR γ) and peroxisome proliferator-activated receptor α (PPAR α), and another regulator of the adipogenesis (C/EBP α) which play important role in inducing the expression of PPAR γ and C/EBP α , thereby causing adipogenesis. Peroxisome proliferator-activated receptor gamma (PPAR- γ) is mainly expressed in adipocytes and regulates lipid and glucose metabolism. It is essential for adipocyte proliferation and differentiation and plays an important role in enhancing anti-inflammation and insulin sensitivity by promoting adiponectin expression in various peripheral tissues [7]. Moreover, obesity is also associated with an increase in the size and amount of adipose tissue [8]. Adipose tissue secretes a number of signaling molecules known as adipokines including leptin and adiponectin [9]. Additionally, the lipoproteins lipase (LPL) enzyme, sterol regulatory element-binding proteins (SREBPs), adipocyte fatty acid binding protein (aP2), and fatty acid synthase (FAS) are also the key regulatory factors involved in adipose tissue metabolism and fatty acid oxidation [10–12].

Current medications used to treat obesity were found to have limited efficacy and undesirable side effects [13,14]. Research has thus focused on developing therapeutic agents able to regulate energy metabolism without affecting other systems and pathways. Some natural products including curcumin, resveratrol, and berberine are known to present anti-obesity effects [13]. In addition, the potential usefulness of whole fruit in preventing overweight and obesity is related to their low energy density, enhanced water content, and enrichment with dietary fibers, mainly comprised of viscous dietary fibers [15].

Pear is the common name for approximately 20 species of trees belonging to the genus *Pyrus* in the rose family (*Rosaceae*), which have been used as traditional medicine in China for over 2000 years based on their anti-inflammatory, anti-hyperglycemic, diuretic, and antitussive effects. Among the *Pyrus* species, *Pyrus ussuriensis* Maxim, also called the “Asian, Korean, or Oriental” pear, is a good source of flavonoids such as cyanidin, catechin and epicatechin and phenolic acids, particularly chlorogenic acid, arbutin, ferulic and citric acid [16,17]. A previous review on the benefits of chlorogenic acid identified 27 studies including three human clinical trials and 18 animal studies showing improved metabolic biomarkers related to diabetes, cardiovascular health and obesity (e.g., serum lipids, insulin resistance, glucose tolerance, and obesity-related hormones) [18]. Moreover, the glycemic index (GI) of whole pears ranges from 33 to 42 (average of 38), based on four studies involving participants with type 2 diabetes mellitus (T2DM) or impaired glucose tolerance (IGT) [19].

Recently, the anti-obesity effect of probiotics has attracted public attention, revealing the potential of gut microbiota in addressing high serum cholesterol levels. *Lactobacilli* are one of the vital health-enhancing species of microbiota, frequently found as probiotics in cultured milk, infant foods, and different pharmaceutical preparations. A beneficial effect suggested to result from human consumption of probiotic lactic acid bacteria (LAB) is a reduction in serum cholesterol levels [20]. One of the LAB probiotics, *Lactobacillus plantarum*, is also reported to exert hypo-cholesterolemic effects in different studies [21]. It was demonstrated in a previous study that *Bacillus subtilis* has the hypoglycemic and antilipidemic activities [22]. Several studies have further reported hypolipidemic effects of probiotic-fermented food [23]. Additional beneficial effects of probiotics or food products containing

probiotics on bodyweight control and obesity related metabolic disorders were also recently revealed, although the relationship between probiotics and obesity remains largely elusive [24–26]. In this study, we aimed to elucidate the outcomes and underlying molecular mechanisms of *Pyrus ussuriensis* Maxim extracts (PUE), both fermented (FPUE) and non-fermented (NFPUE), in a context of HFD-induced obesity using *in vitro* and *in vivo* experimental models.

2. Materials and methods

2.1. Chemicals and reagents

Fenofibrate, dexamethasone, 3-isobutyl-1-methylxanthine (IBMX), and Oil Red O (ORO) powder were purchased from Sigma-Aldrich (St. Louis, MO, USA), while penicillin and streptomycin were from Gibco Life Science Technology (NY, USA).

2.2. Extract preparation, fermentation, and analysis

2.2.1. Preparation of NFPUE and FPUE

The NFPUE and FPUE were prepared followed previously reported methods with slight modifications [27–29]. Briefly, the whole fruit of *Pyrus ussuriensis* Maxim was macerated with water (100 g in 1 L of water) for 8 h at 100 °C. The novel functional FPUE test product was prepared by co-fermenting 50 mL of the concentrated PUE with probiotics. Briefly, the extract was sterilized (15 min at 121 °C) with glucose (3 %) and monosodium L-glutamic acid (5 %). Then, *Bacillus subtilis* HA (KCCM 10775P) starter culture was inoculated and incubated at 42 °C for three days. Then, the product was mixed with skim milk (1 %, v/v) and glucose (1.5 %, v/v) solutions. *Lactobacillus plantarum* EJ2014 (KCCM 11545P) was inoculated and incubated at 30 °C over seven days for secondary fermentation. Finally, the fermented product was lyophilized using a freeze dryer (Ilshin BioBase Ltd., Ede, the Netherlands).

2.2.2. Phytochemical analysis

Gas Chromatography Coupled Mass Spectroscopy (GC-MS) and Liquid Chromatography/Mass Spectrometry (LC/MS) analysis were performed to elucidate the phytochemical profile of NFPUE and FPUE. Briefly, a 5975 GC-MS selective detector (Agilent Technologies, CA, USA) was used an ionization energy of 70 eV, scanning mass of 10–800 *m/z*, and run time of 67 min for GC/MS analysis. An Accela UHPLC system (Thermo Fisher Scientific, CA, USA) coupled with an LTQ-Orbitrap XL hybrid mass spectrometer (Thermo Electron, Bremen, Germany) via an ESI interface was used for LC/MS analysis [30].

2.3. In vitro outcomes of NFPUE and FPUE on 3T3-L1 adipocytes differentiation

2.3.1. Cell culture

Murine preadipocyte 3T3-L1 cells were purchased from the Korean Cell Line Bank (Seoul, Republic of Korea). The American Type Culture Collection (ATCC) guidelines were strictly followed for cell culture, maintenance, viability, differentiation, and post-differentiation [14,31]. Briefly, serum and media were purchased from Sigma-Aldrich (St. Louis, MO). The 3T3-L1 cells were cultured in their growth media (GM) constituted as Dulbecco's modified Eagle's medium (DMEM) supplemented with 10 % bovine calf serum (BCS) and 1 % Penicillin (100 IU/mL) as well as Streptomycin (10 mg/mL) until confluence.

2.3.2. Differentiation of 3T3-L1 preadipocytes

The treatment and analysis performed to evaluate the effects of NFPUE and FPUE at different concentrations (50,100, and 500 μ g/mL) on the differentiation of 3T3 cells into adipocytes are summarized in Fig. 1A. Briefly, the confluent preadipocyte 3T3-L1 cells were subjected to differentiation with differentiation media (DM) including DMEM containing 10 % BCS, IBMX (500 μ M), dexamethasone (1 μ M) and

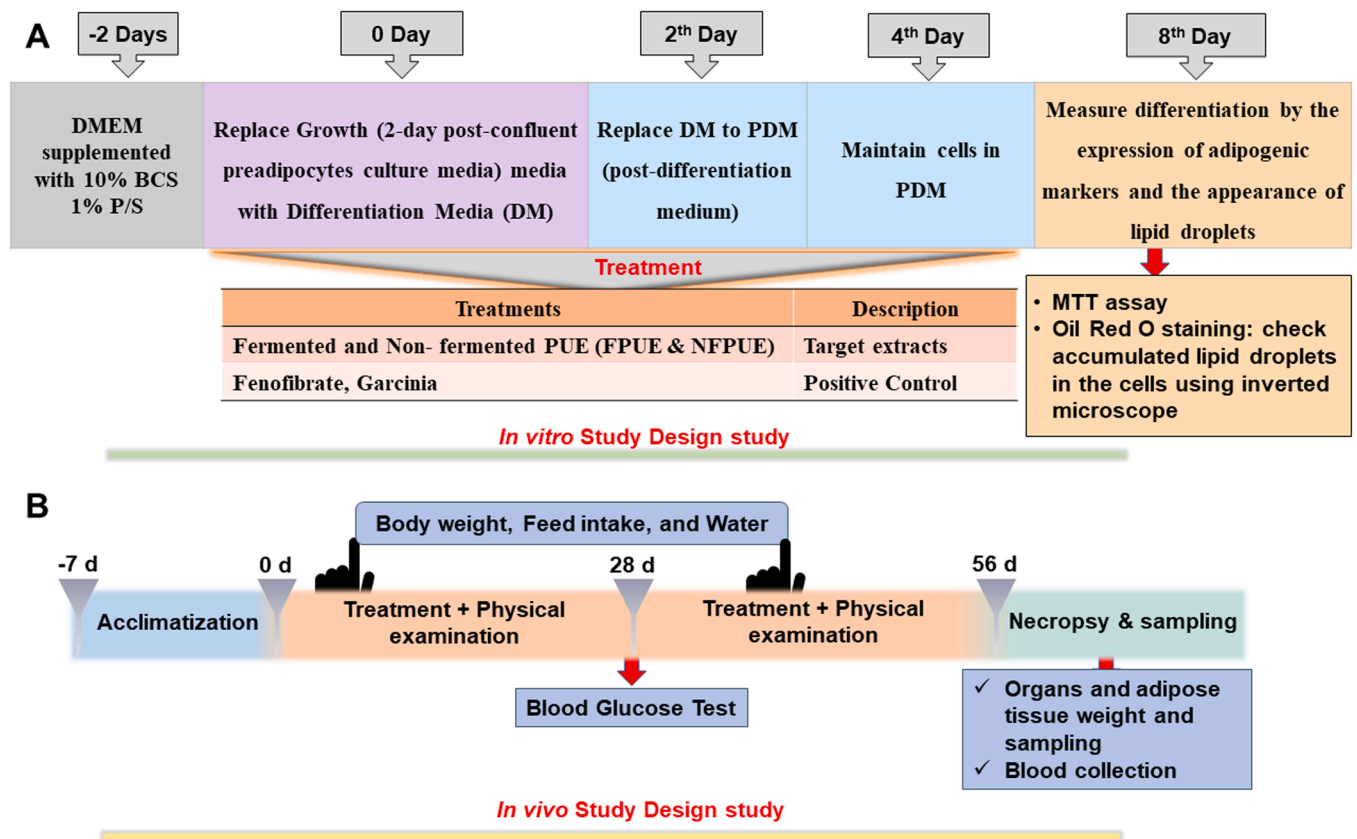


Fig. 1. (A) *In vitro*, and (B) *In vivo* study design to evaluate the lipid accumulation and obesity-associated effects of fermented and non-fermented *Pyrus ussuriensis* Maxim extract using 3T3-L1 preadipocyte cells and High-fat diet-induced obesity model of rats, respectively.

insulin (5 µg/mL) (Day 2). After 2 days, the DM was replaced by post-differentiation media (PDM) including DMEM supplemented with 10 % BCS and insulin (5 µg/mL) (Day 4). Then, the PDM was replaced by DMEM containing 10 % BCS without insulin for four additional days (Day 8). As positive controls, Fenofibrate (FF) and Garcinia (HCA) were used instead of NFPUE and FPUE.

2.3.3. Oil Red O (ORO) staining

Differentiated 3T3-L1 cells were fixed with 10 % formalin and washed for 10 min with phosphate-buffered saline. Then, the cells were stained with ORO (0.3 %). After washing with distilled water, the cells were examined under a microscope. Finally, the ORO stain bound to the cells was eluted with isopropanol (100 %) and quantified using the optical density measured at 590 nm (OD₅₉₀).

2.3.4. Assessment of cell viability using a colorimetric assay

Cytotoxicity of NFPUE and FPUE was determined by assessing the mitochondrial-dependent reduction of 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) to formazan with a rapid colorimetric assay. Briefly, the 3T3-L1 cells were plated in 96-well plates and incubated until confluence. The confluent 3T3-L1 cells were then treated with different concentrations (1–1000 µg/mL) of NFPUE and FPUE for the indicated time. Subsequently, the optical density (OD) of the culture wells was determined using a microplate reader measuring absorbance at 540 nm. The cell viability formula used is the following:

$$\text{Cell viability (\%)} = (\text{OD value of sample} / \text{OD value of control}) \times 100$$

2.4. In vivo study design and sample handling

2.4.1. Diet

The high-fat diet (HFD) was prepared based on the composition provided by Levin and Dunn-Meynell [32]. The normal rat chow diet contains 3.06 kcal/ g consisting of 3 % fat, 48.8 % carbohydrate, and 21 % protein. The HFD [Adjusted Calories Diet (45/FAT) TD-06415] contains 4.16 kcal/ g consisting of 44.8 % fat, 36.2 % carbohydrate and 19 % as protein. The composition of the HFD and normal chow diet is provided in Table 1.

Table 1

Composition of high-fat diet and normal chow diet for rats.

High-fat diet		Normal rat chow diet	
Nutrients	%	Nutrients	%
	100 g		100 g
Carbohydrate	36.2	Carbohydrate	48.8
Protein	19	Protein	21
Fats	44.8	Fats	3
Ingredients	g/kg	Calcium	0.8
Casein	245.0	Phosphorus	0.4
L-Cystine	3.5	Fiber	5
Corn Starch	85.0	Moisture	13
Maltodextrin	115.0	Ash	8
Sucrose	200.0		
Lard	195.0		
Soybean Oil	30.0		
Cellulose	58.0		
Mineral Mix, AIN-93G-MX (94046)	43.0		
Calcium Phosphate, dibasic	3.4		
Vitamin Mix, AIN-93-VX (94047)	19.0		
Choline bitartrate	3.0		
Red Food Color	0.1		
Total energy (kcal/g)	4.16	Total energy (kcal/g)	3.062

2.4.2. Animal treatment

The experiment was conducted according to the international guidelines for the care and use of laboratory animals [33] and approved by the Animal Care and Use Committee of Kyungpook National University, Daegu, Korea (Approval no.: KNU 2019-0072). Fifty-six albino Sprague Dawley rats (4-week-old males) were purchased from Orient Co. (Gyeonggi-do, Korea). The total number of rats ($n = 56$ animals) used in the present study was calculated using power analysis with G*power based on an effect size of 0.5, α -error probability of 0.05, power ($1-\beta$ error probability) of 0.6, and the number of groups of seven [34]. All rats were housed with a 12 h light-dark cycle and free access to water and food *ad libitum*. After acclimation for 1 week, the rats were randomized into seven groups ($n = 8$ animals/group) and treated for 8 weeks with a normal chow diet or HFD (45 %) as shown in Fig. 1B. Food intake and body weight were monitored daily. The daily dose intake is provided in Table 2. At the end of the experiment, the feed conversion ratio (FCR) was calculated from the total consumption of food divided by the total body weight gain over 8 weeks. In the 4th week of treatment, fasting blood glucose level was determined for each group of rats using an ACCU-Check® Active blood glucose meter (Roche Diabetes Care GmbH, Altek Corp. Ltd). After 8 weeks, the rats were fasted for 16 h and euthanized using CO₂. Blood was collected into heparinized tubes, plasma was separated and stored at -80°C until use. Organs were harvested and frozen immediately in dry ice and stored at -80°C until use.

2.4.3. Measurement of obesity-associated biomarkers in plasma

The plasma concentration of the incretins leptin and ghrelin, as well as adiponectin, insulin, gastric inhibitory peptides (GIP), gamma-glutamyl transferase (GGT), low-density lipoprotein (LDL), high-density lipoprotein (HDL), triglycerides (TG), and free fatty acids (FFA) were determined using enzyme-linked immunosorbent assay kits (Sigma-Aldrich, St. Louis, MO, USA; Abcam Cambridge, UK) following the manufacturer's instructions. Furthermore, the plasma levels of TNF- α and IL-6 were determined to evaluate the anti-inflammatory activities of the extracts.

2.4.4. Morphometric analysis of tissues

The liver, epididymal, and mesenteric adipose tissue were fixed in 10 % buffered formalin for 24–28 h at 4°C , processed for paraffin embedding, and stained with hematoxylin and eosin (H & E). Cell size and lipid cells were analyzed under a light microscope (10×200) for morphometric analysis.

2.4.5. RNA isolation and quantitative reverse transcription-polymerase chain reaction (qRT-PCR) analysis

For RT-PCR experiments, total RNA was extracted from the tissue samples using a RNeasy reagent (Invitrogen, CA, USA) according to the manufacturer's instructions. The resulting solution was diluted 100-fold, purity was confirmed, then cDNA (1 μg) was synthesized from RNA using an Eco Dry Premix (Oligo dT) (Clontech Laboratories, Inc., Seoul, South Korea). The Real-time RT-PCR was performed using a

SYBR® Green master mix and thermocycler, both of which were purchased from Applied Biosystems (Foster City, CA, USA). The primers used are listed in Table S1.

2.5. Microbiota isolation and analysis

Microbial DNA was extracted and purified from fecal samples using a QIAamp DNA Stool Mini Kit (QIAGEN Inc., Hilden, Germany) according to the manufacturer's instructions. The Quality control requirement of the DNA sample was met by achieving a DNA concentration of ≥ 15 ng/ μL , and A_{260}/A_{280} ratio of ≥ 1.8 . Microbiome profiling was conducted based on 16 S rRNA gene sequences obtained from the rats' fecal sample and amplified using the universal bacterial PCR primers 27F (AGRGTITGATYMTGGCTCAG) and 1492R (GGYTACCTTGTACGACTT) with a Pacific Biosciences (PacBio) RS1 system from ChunLab Inc. (Seoul, Republic of Korea). Microbiome Taxonomic Profile (MTP) was generated using next-generation sequencing (NGS) with the EzBioCloud MTP pipeline and EzBioCloud 16S database PKSSU4.0. Sequences were separated by unique in-house codes. Low-quality reads including sequences with read lengths of 100–2000 bp, average Q values < 25 , or not matching any of reference sequences with a ≤ 97 % similarity cut-off clustered using UCLUST were filtered out. The species identified in the query sequences achieved 97 % similarity with the EzBioCloud 16S database, while the OTUs obtained by those not achieving 97 % similarity with the database (which then clustered with the UCLUST tool) was used to generate a final set of OTUs for alpha diversity indices (including Shannon and Chao1 indices), taxonomic composition, and relative abundance of microbiota using EzBioCloud 16S-based MTP application [35].

2.6. Statistical analysis

Data were expressed as mean $\pm$ standard error of the mean (SEM). We employed a one-way analysis of variance (ANOVA) followed by Dunnett's multiple range tests to compare between groups for all measurements except the FCR using Prism 5.01 (Graph-Pad, La Jolla, CA, USA). Non-linear regression analysis (Prism 5.01 program, GraphPad Software, Inc., U.S.A) was used to obtain 50 % inhibitory concentration (IC₅₀) values. A threshold of $P < 0.05$ was considered statistically significant.

3. Results

3.1. Phytochemical profile of NFPUE and FPUE

The LC-MS and GC-MS analyses of NFPUE and FPUE investigated numerous types of compounds. The main compounds identified, and their respective peaks based on the GC/MS and LC/MS analysis are described in Figs. 2 and 3, respectively. Particularly, catechol, erythritol, and L-arabinitol were identified as the major compounds of NFPUE, while 3-butanediol, lactic acid, b-hydroxyphenyl lactic acid, and tetramethyl pyrazine were identified as the principal compounds of FPUE.

3.2. In vitro effects of NFPUE and FPUE on 3T3-L1 adipocytes differentiation

3.2.1. NFPUE and FPUE are not cytotoxic

To assess the cytotoxicity of NFPUE and FPUE on 3T3-L1 cells, their viability after exposure to a series of concentrations of NFPUE and FPUE was determined using the MTT assay. As shown in Fig. 4A, viability decreased up to 16.2 % with NFPUE and to 14.3 % with FPUE, at the highest tested concentration of 1 mg/mL. These results suggest that the fermented extract of *Pyrus ussuriensis* Maxim is safer compared to its non-fermented form based on increased cellular viability.

Table 2

Scheme of study and daily dose (per kg body weight) intake for each group of rats.

Groups		Daily intake
ND	Normal Diet	Normal feed
HFD	High-fat Diet	Feed with High-fat Diet (HFD)
HCA	Positive Control	500 mg/kg b.w HCA (Garcinia Extract) + HFD
FPUE1	Dose 1	250 mg/kg b.w of fermented PUE + HFD
FPUE 2	Dose 2	500 mg/kg b.w of fermented PUE + HFD
FPUE 3	Dose 3	1 g/kg b.w of fermented PUE + HFD
NFPUE	Non-fermented extract	500 mg/kg b.w of Non-fermented PUE + HFD

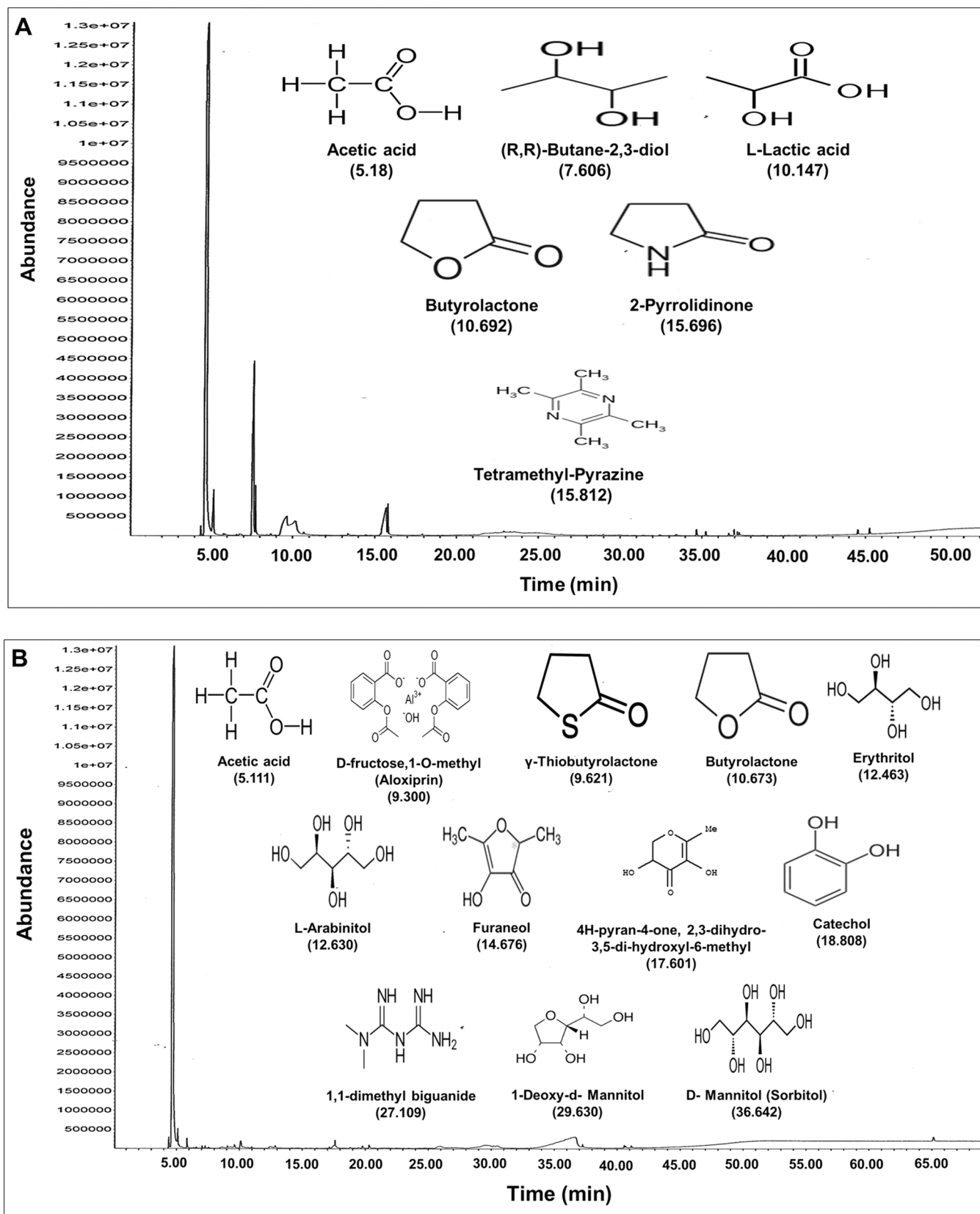


Fig. 2. GC/MS analysis of the fermented (A) and non-fermented (B) *Pyrus ussuriensis* extract (PUE). In the chromatograms generated from the GC/MS analysis of FPUE and NFPUE, each peak represents a specific compound obtained at a specific retention time (RT). The chemical structures of the bioactive compounds together with their RT are also provided.

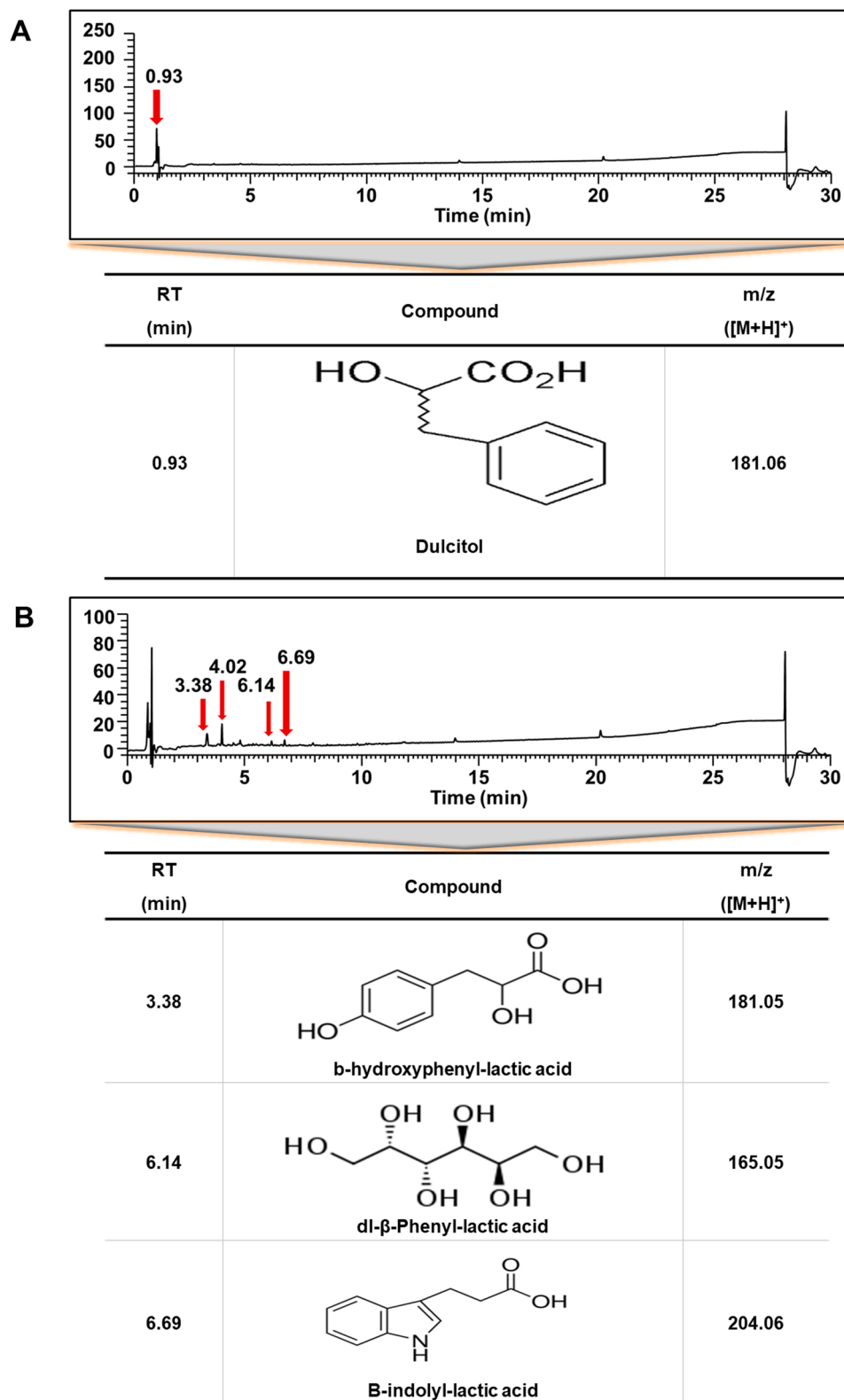


Fig. 3. LC/MS analysis of non-fermented (A) and fermented (B) *Pyrus ussuriensis* extract (PUE). In the chromatograms generated from the LC/MS analysis of FPUE and NFPUE, each peak represents a specific compound obtained at a specific retention time (RT). The chemical structures of the bioactive compounds together with their RT and *m/z* value are also provided.

3.2.2. NFPUE and FPUE inhibit adipogenesis

To evaluate the anti-adipogenic effects of NFPUE and FPUE, pre-adipocyte 3T3-L1 cells were fully differentiated into mature adipocytes in the presence of NFPUE and FPUE. Their lipid accumulation was

measured using Oil Red O staining on day 8 following NFPUE and FPUE administration. As shown in Fig. 4B, the IC₅₀ values of NFPUE and FPUE were 96.67 µg/mL and 85.33 µg/mL, respectively. For FPUE, the IC₅₀ value was lower (5.8 %) than HCA (90.60 µg/mL) and higher (5.3 %) than

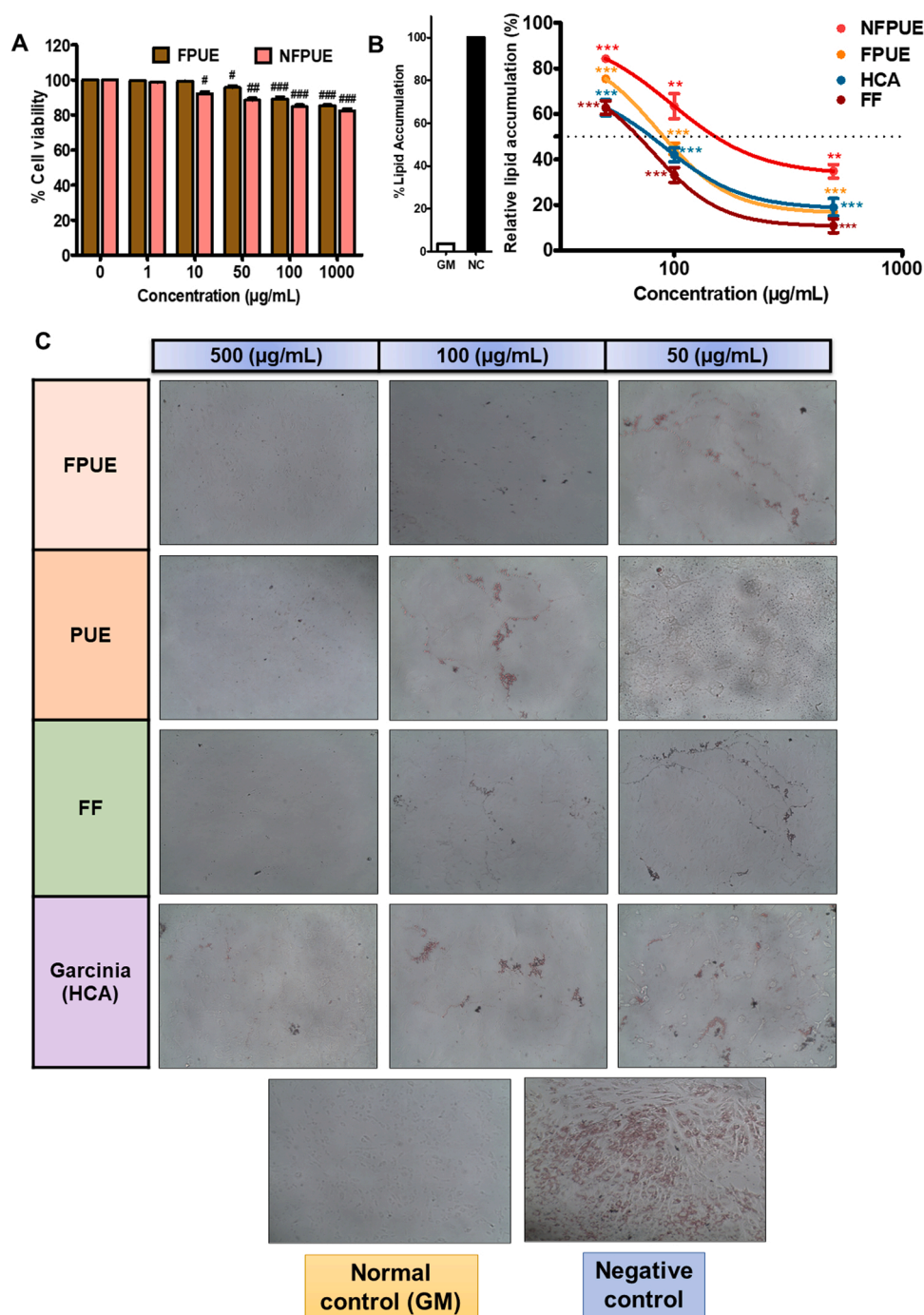


Fig. 4. *In vitro* effects of FPUE and NFPUE in 3T3-L1 adipocytes. In the murine 3T3-L1 pre-adipocyte cell line, (A) the cytotoxicity of FPUE and NFPUE was evaluated using a cell viability assay, while (B) the accumulation of lipids was measured using solubilized lipid droplets extracted from the Oil Red O-stained cells with a colorimetric assay. The absorbance of the solution was measured at 510 nm. However, the growth media (GM) was used as normal control. (C) Microscopic images captured at $200\times$ magnification showing the accumulation of lipids are provided. Garcinia and fenofibrate were used as positive controls. The values are expressed as mean $\pm$ standard error of the mean (SEM) for three independent experiments. * ($p \leq 0.05$), ** ($p \leq 0.01$), and *** ($p \leq 0.001$), different from negative control (NC) group, # ($p \leq 0.05$), ## ($p \leq 0.01$), and ### ($p \leq 0.001$), different from normal control group.

than fenofibrate (80.75 $\mu\text{g/mL}$). In the case of NFPUE, it was one time (11.7 %) greater than the non-fermented extract. These results suggest that fermentation enhances the inhibitory effects of PUE against adipogenesis. Microscopic examination of ORO-stained cells further revealed that both the non-fermented and fermented extracts significantly ($p < 0.05$) reduced lipid accumulation at various concentrations (50, 100 and 500 $\mu\text{g/mL}$) compared to the non-treated control (GM) (Fig. 4C). However, the reduction was more pronounced in the FPUE versus NFPUE treated cells, resulting in a percent lipid accumulation of 16.87 % and 34.7 %, respectively, with a concentration of 50 $\mu\text{g/mL}$ (Fig. 4B).

3.3. *In vivo* effects of NFPUE and FPUE in HFD-induced obese rats

3.3.1. NFPUE and FPUE reduce body weight gain in HFD-induced obese rats

To examine the effects of NFPUE and FPUE on body weight gain, we measured the body weight of rats from all the groups across the experimental period. There was an increase in body weight over time among all groups, but the trend varied between the groups. The HFD-fed rats gained more weight compared to the other groups within the experimental period (Table 3 and Fig. 5A). However, a significant ($p < 0.05$) reduction of 21.16 % in body weight gain was observed for the FPUE-treated obese rats versus non-treated HFD-induced obese rats. A similar trend was observed for the morphology of the rats' bodies (Fig. 5B). Additionally, food intake varied slightly across the different

Table 3

Effects of fermented *Pyrus ussuriensis* Maxim extracts at 250 mg (FPUE1), 500 mg (FPUE2) and 1000 mg (FPUE3) per kg of body weight or non-fermented *Pyrus ussuriensis* Maxim extracts at 500 mg/kg of body weight on food intake, FCR and blood glucose (g/dL).

Parameters	ND	HFD	HCA	FPUE1	FPUE2	FPUE3	NFPUE
Body weight gain (g/rat)	381.6 ± 3.2***	560 ± 4.3	473.1 ± 2.8*	458.8 ± 5.2**	445.9 ± 6.6***	441.5 ± 2.9***	519.9 ± 7.8
Feed intake (g/rat)	120.0 ± 2.0**	98.6 ± 1.0	95.0 ± 1.7	87.9 ± 0.4	90.1 ± 7.3	92.4 ± 3.9	103.5 ± 0.8
FCR ^A	31.85	17.62	20.31	19.41	20.48	21.20	20.16
Glucose (g/dL)	94.7 ± 4.8**	120.6 ± 2.5	99.7 ± 7.1*	99.8 ± 4.9*	102.8 ± 3.9	98.2 ± 6.1*	100.8 ± 3.2*
Triglyceride (mg/mL)	180.1 ± 5.4***	560.6 ± 8.0	254.9 ± 4.3***	520.8 ± 5.1**	291.5 ± 7.3***	188.9 ± 6.8***	324.1 ± 8.5***
AI ^B	45.1 ± 8.3*	95.0 ± 2.1	34.9 ± 13.9**	52.2 ± 10.8	40.1 ± 4.2*	23.8 ± 5.0**	52.5 ± 9.4

Values are expressed as mean ± SEM, (n = 8).

* ($p \leq 0.05$), different from HFD group.

** ($p \leq 0.01$), different from HFD group.

*** ($p \leq 0.001$), different from HFD group.

^A Feed conversion rate was calculated from the total consumption of food divided by the total body weight gain in eight rats over eight weeks; analysis of variance was not used for the analysis of FCR.

^B Atherogenic Index (AI) = (TC – HDL)/HDL.

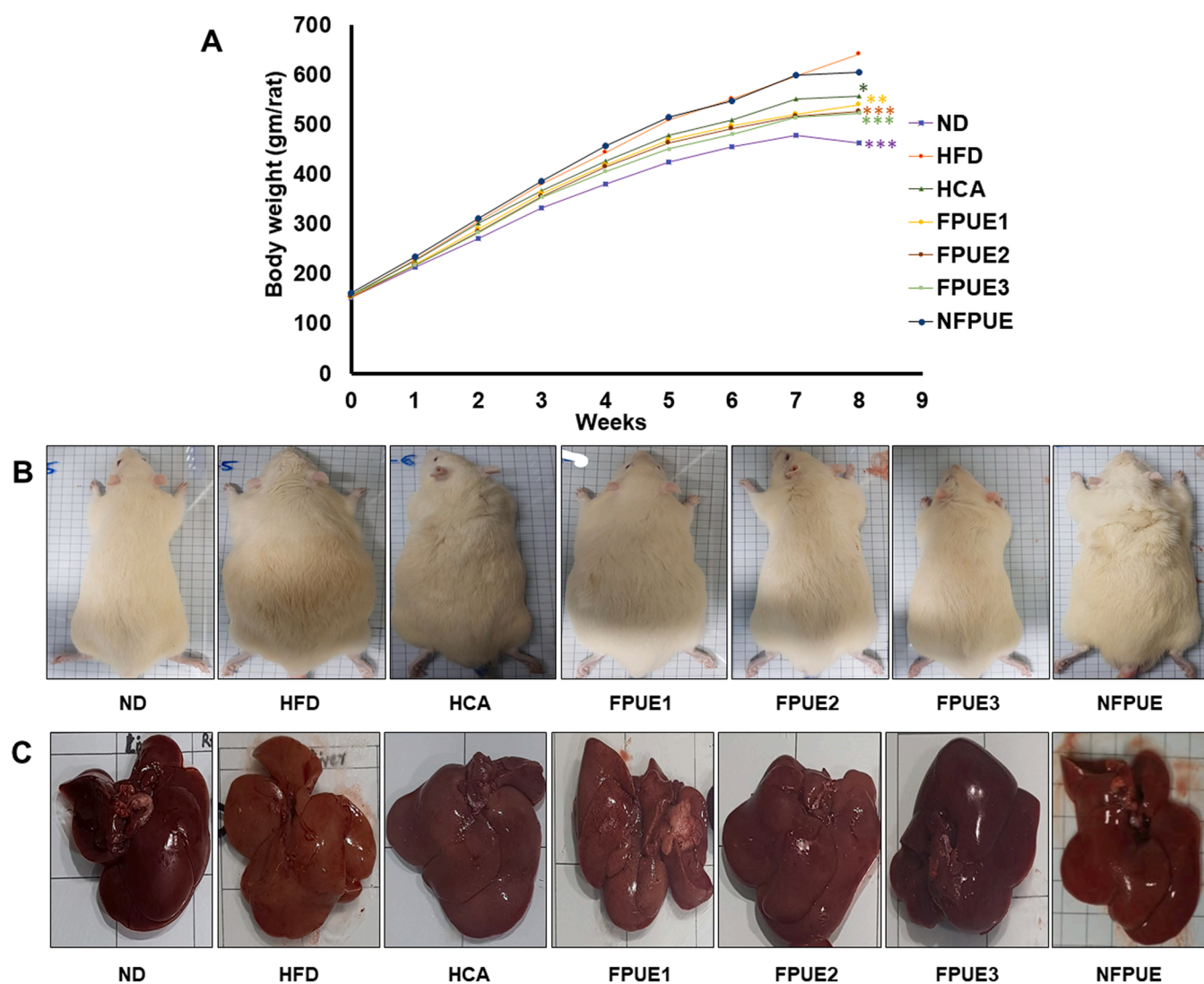


Fig. 5. Effects of FPUE and NPUE on body weight and gross morphology of rats. The figure shows (A) body weight changes per week; (B) gross morphology of rats and (C) their liver from different groups including normal diet group (ND), a high-fat diet group (HFD), or groups fed with FPUE-HFD at different doses of 250 mg (FPUE1), 500 mg (FPUE2), 1000 mg (FPUE 3) per kg b.w and NFPUE at the dose of 500 mg/kg b.w (NFPUE). The image of the liver of a rat from the HFD group shows a pale color and spotted surface as compared to a normal reddish-brown color liver from the ND group. The morphological characteristics of organs from treatment groups are similar to the ND group. Values are expressed as mean ± SEM (n = 8). * ($p \leq 0.05$), ** ($p \leq 0.01$), and *** ($p \leq 0.001$), different from HFD group.

groups of rats (Table 3). The FCR was 16.8 % higher for the FPUE-treated obese rats compared to the HFD-fed rats. As shown in Table 3, the body weight gain reduction under FPUE treatment was 6.6 % higher compared to HCA treatment, which suggested that FPUE could prevent body weight gain better than HCA under HFD.

3.3.2. NFPUE and FPUE reduce fat mass in HFD-induced obese rats

As shown in Fig. S1A and B, there was a significant difference ($p < 0.05$) in the weight of the kidney, and spleen among the different groups. The weight of liver and adipose tissues (subcutaneous, mesenteric, perineal, and epididymal) also increased considerably in HFD-fed rats compared to control chow rats (Fig. S1C–G). Furthermore, the

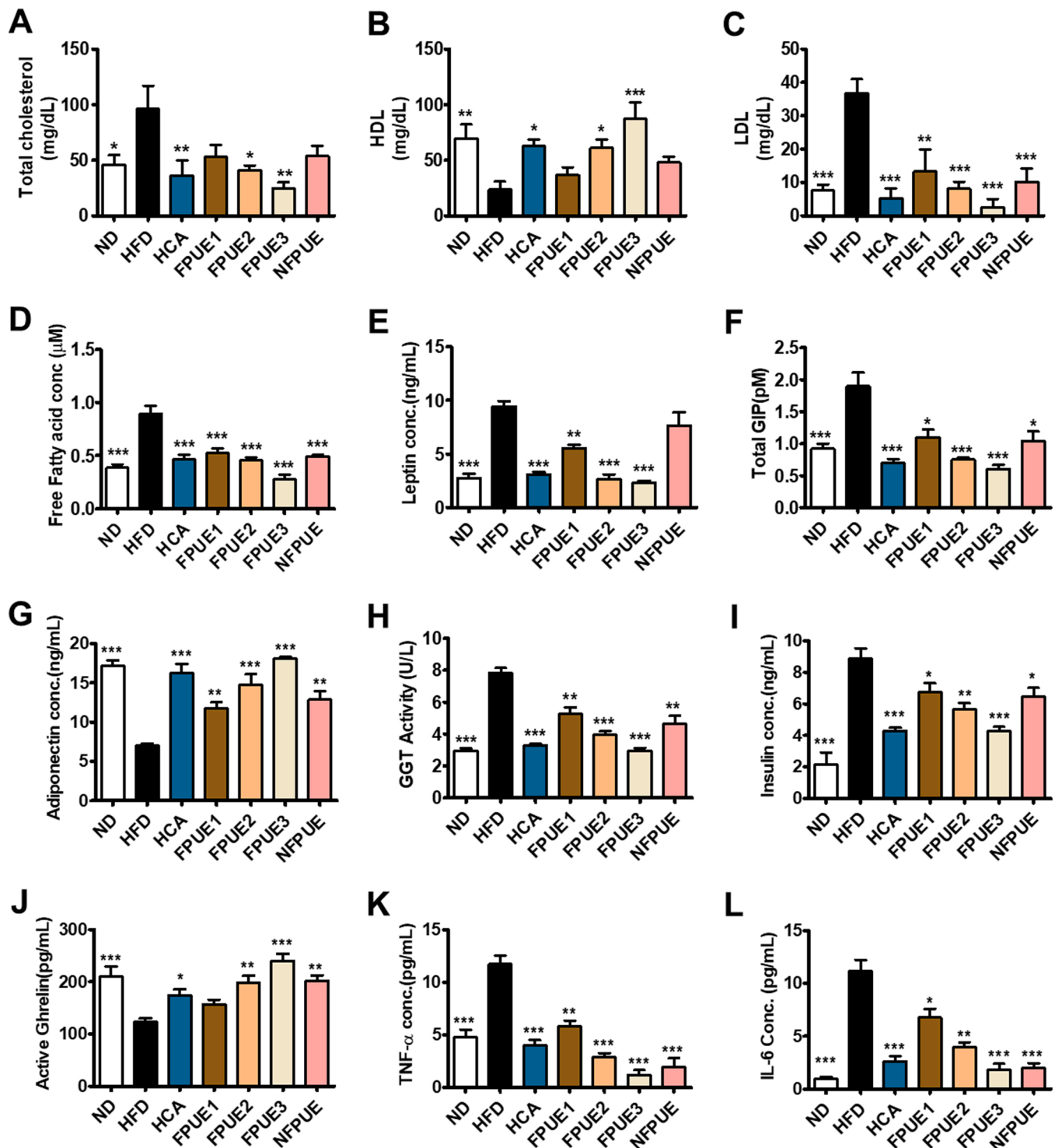


Fig. 6. *In vivo* effects of FPUE and NFPUE on liver function biomarkers in plasma. The lipid profile (TC, HDL, and LDL), pro-inflammatory cytokines, endocrine peptides, incretins, and liver function enzymes in plasma following treatment with 250, 500, and 1000 mg/kg of body weight of FPUE and 500 mg/kg of NFPUE in rats receiving a high-fat diet (HFD) for 8 weeks. (A) Total cholesterol; (B) HDL (High Density Lipoproteins); (C) LDL (Low Density Lipoproteins); (D) FFA (Free Fatty acid); (E) Leptin; (F) Total GIP; (G) Adiponectin; (H) Gama-Glutamyl Transferase (GGT) activity; (I) Insulin; (J) Active Ghrelin; (K) TNF-α; and (L) IL-6. Values are expressed as mean ± SEM (n = 8). * ($p < 0.05$), ** ($p < 0.01$), and *** ($p < 0.001$), different from HFD group.

weight of liver and adipose tissues was noticeably reduced in the FPUE-treated *versus* non-treated HFD-fed rats. Moreover, as shown in Fig. 5C, the liver of HFD group rats was pale in color with a spotted surface, compared to a normal reddish-brown color from the control chow group.

3.3.3. NFPUE and FPUE regulate blood glucose in HFD-induced obese rats

The blood glucose levels were significantly ($p < 0.05$) reduced in the NFPUE- and FPUE- treated *versus* non-treated HFD groups, after four weeks of treatment, irrespective of the dose of extract administered (Table 3).

3.3.4. NFPUE and FPUE regulate liver function biomarkers in the plasma of HFD-induced obese rats

The impact of NFPUE and FPUE on the plasma lipid profile is presented in Fig. 6A–D. Elevated concentrations of TC, LDL, and FFA were observed, providing evidence of hyperlipidemia in the HFD-fed obese rats. Additionally, treatment with FPUE (250–1000 mg/kg of body weight) significantly ($p < 0.05$) reduced the plasma levels of these markers in the HFD-fed rats. Similarly, FPUE treatment significantly ($p < 0.05$) lowered the atherogenic index (AI) values in HFD-fed rats (Table 3).

The impact of FPUE on the plasma levels of leptin, total GIP, adiponectin, GGT, insulin, ghrelin, TNF- α , and IL-6 in the HFD-fed rats is presented in Fig. 6E–L. Receiving a HFD increased plasma levels of TNF- α , IL-6, and leptin, which were significantly ($p < 0.05$) reduced upon FPUE treatment (250–1000 mg/kg) in the HFD-fed rats (Figs. 6E, 6K, and 6L). The plasma levels of total GIP, gamma-glutamyl transferase, and insulin were also reduced significantly ($p < 0.05$) with the FPUE-supplemented HFD (Fig. 6F, 6H, and 6I). Moreover, the attenuation of adiponectin, and ghrelin resulting from HFD intake was significantly ($p < 0.05$) prevented by the FPUE treatment depending on the dose (Fig. 6G and 6J).

3.3.5. NFPUE and FPUE reduce hepatocyte and adipocyte size in HFD-induced obese rats

The histopathological alterations observed in liver tissues of rats receiving a HFD with NFPUE and FPUE are summarized in Fig. 7A. By contrast, the liver slices of control chow-fed rats exhibited a normal lobular architecture with well-structured hepatocytes. Receiving the HFD for 8 weeks induced fat droplet accumulation in the hepatic tissues of rats. A substantial increase in the accumulation of fat droplets was observed in the livers of the HFD-fed rats compared to control chow-fed rats. The supplementation with NFPUE and FPUE resulted in less accumulation or an absence of fat droplets in the liver tissues of HFD-fed rats. The architecture of hepatocytes and the morphology of hepatic tissues of NFPUE and FPUE-treated rats were also similar to control chow-fed rats.

The effects of FPUE on the adipocyte size of epididymal and mesenteric adipose tissues are shown in Figs. 7B and 7C, respectively. The size of adipocytes in HFD-fed rats was markedly increased compared to control chow-fed rats. By contrast, supplementation with FPUE reduced the size of adipocytes in the adipose tissues of HFD-fed rats.

3.3.6. NFPUE and FPUE regulate lipogenic and adipogenic genes expression in HFD-induced obese rats

The mRNA expression of lipogenic genes, PPAR γ , C/EBP α , and SREBP-1 C was significantly up-regulated in HFD-fed obese rats compared to control chow-fed rats. These genes were further down-regulated in rats treated with FPUE. The mRNA expression of FAS and aP2 genes was also significantly downregulated following FPUE treatment (Fig. 8). Moreover, the mRNA expression of ACC1, LPL, and GLUT4 were downregulated in HFD-fed rats compared to control chow-fed rats. The expression of the GLUT4 gene was normal following FPUE supplementation in HFD-fed rats. However, FPUE treatment did not have significant effects on ACC1 and LPL gene expression.

3.4. FPUE modulates the intestinal microbiota in HFD-induced obese rats

The intestinal microbiota can prevent obesity by regulating metabolic pathways. High-throughput 16 S rRNA gene sequencing yielded 632,533 good-quality sequences from nine HFD-induced obese rat samples with and without FPUE treatment. We detected nine bacterial phyla and 361 genera among these samples, which revealed that the gut microbiota structure and composition were significantly influenced by the HFD. Firmicutes was the most abundant phyla at all taxonomic levels, while *Oscillibacter*, *Escherichia*, and *Klebsiella* were the most abundant genera among the OTUs (Figs. S2 and S3). The FPUE treatment significantly reversed this bacterial abundance in the HFD rats, similar to the control chow group. The taxonomic profiles further indicated that the proportion of *Bacteroides*, *Verrucomicrobia*, and *Proteobacteria* decreased significantly, whereas the abundance of Firmicutes increased markedly, in the HFD-fed rats compared to the control chow and FPUE groups. In the FPUE-treated obese rats, *Akkermansia*, *Bacteroides*, *Parabacteroides*, *Lactobacillus*, *Eubacterium*, *Clostridium*, *Bifidobacterium*, *Corynebacterium*, and *Prevotella* were the most abundant genera among the OTUs (Figs. S2, S3 and Fig. 9A). We further analyzed the outcome of FPUE treatment in the HFD-fed rats. Overall valid reads of both groups (HFD and FPUE) were approximately similar to the control chow. However, the OTUs were comparatively higher in HFD-fed rats as compared to the FPUE treatment (Figs. 9B and 9C). Alpha-diversity analysis indicated that the values of Chao1 and Shannon index were significantly ($p < 0.05$) attenuated by HFD compared to control chow. By contrast, FPUE treatment significantly increased these parameters in HFD-fed rats, reaching levels above the control chow group (Figs. 9D and 9E).

4. Discussion

In our search for functional food products presenting anti-obesity activity, we identified the fermentation product of PUE as a potential therapeutic agent. In the present report, we characterized the phytochemical profile and anti-obesity effects of FPUE, as well as investigated the underlying molecular mechanisms. Additionally we have elucidated the outcome of FPUE on microbiota regulation in association with its anti-obesity effects (Fig. 10).

According to the literature, the probiotic *Bacillus subtilis* and *Lactic acid* bacteria can exert a preventive effect on obesity and metabolic syndrome. These probiotic bacteria produce 2,3-butanediol [36–38]. In this study, 2,3-butanediol, lactic acid, b-hydroxyphenyl lactic acid, and tetra-methyl pyrazine were identified as the main compound of FPUE, while catechol, erythritol, and L-arabinol were identified as major compounds of NFPUE. A previous study showed that the b-hydroxyphenyl lactic acid decreased reactive oxygen species production in microglia and neutrophils, acting as a natural anti-oxidant [39]. In addition, dl- β -Phenyl lactic acid and β -indolyl lactic acid are used to reduce antibacterial resistance by biofilm formation and quorum sensing inhibition, respectively, during cancer chemotherapy [40,41]. Erythritol which is used as a sweetening agent is the least sweet among all sugars and can allow for weight control [42]. Butanediol was shown to reduce the cerebral glycolytic rate and improve cerebral energy metabolism, which is closely related to systemic energy metabolism [43,44]. Lactic acid also decreases pro-inflammatory cytokines production, while tetra-methyl pyrazine regulates inflammation and free radical scavenging activity [45,46]. Thus, the anti-obesity effect of FPUE might result from the presence of butanediol, lactic acid, b-hydroxyphenyl lactic acid, and tetramethyl pyrazine in the extract. Additionally, the anti-obesity activity of NFPUE could be due to the presence of catechol, erythritol, and L-arabinol in the extract, as supported by our results and the previous literature.

In this study, we used a HFD-induced obesity rat model considered highly translational because it mimics the main cause of obesity in humans [2]. Moreover, feeding HFD to animals can readily induce body

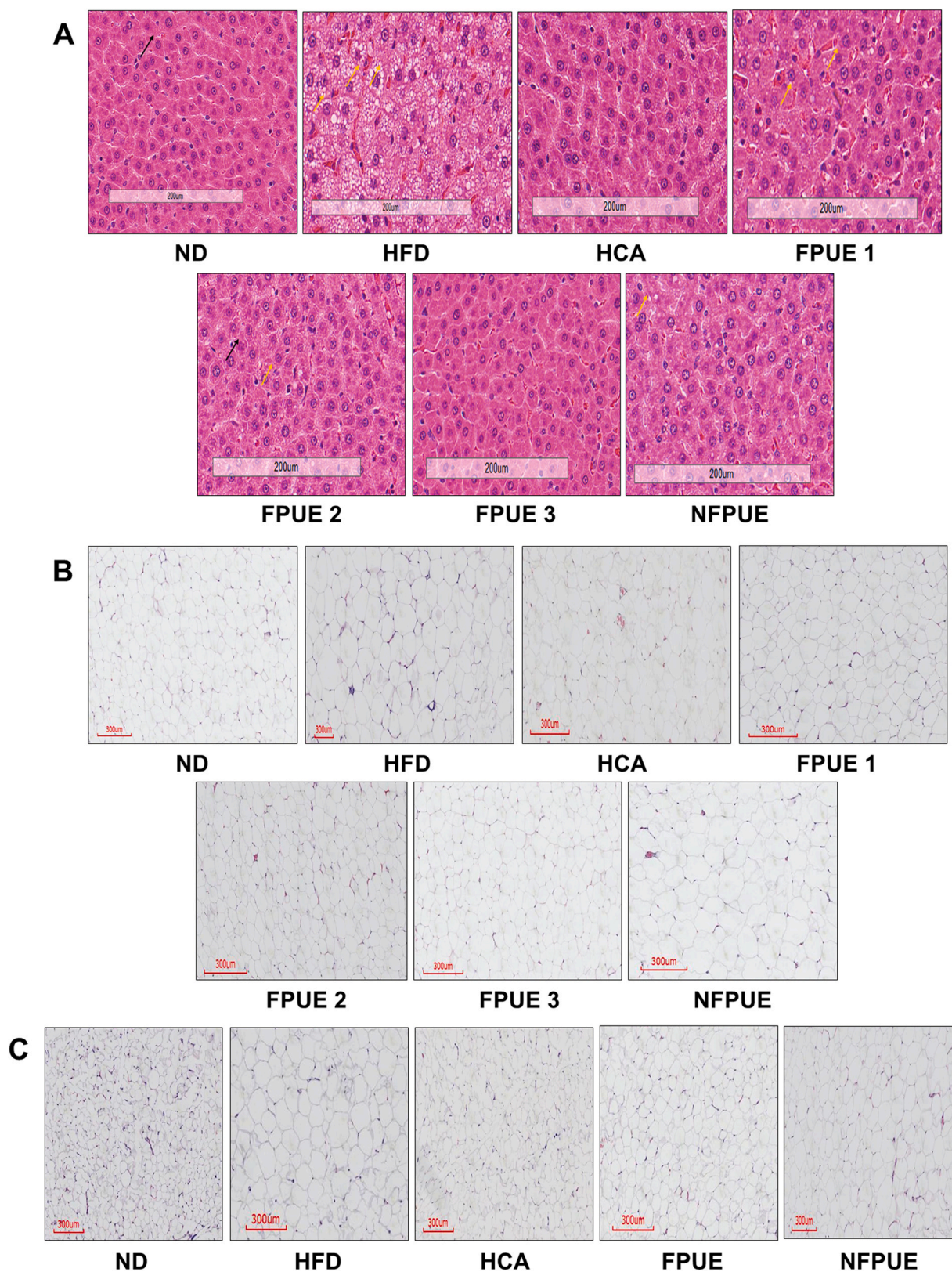


Fig. 7. *In vivo* effects of FPUE and NFPUE on lipid accumulation. Hematoxylin and eosin staining of (A) hepatic, (B) epididymal, and (C) mesenteric adipose tissues (10 × magnification). Rats receiving a normal diet (ND); a high-fat diet (HFD); a HFD accompanied by garcinia extract (HCA) treatment as a positive control; a HFD accompanied by FPUE treatment at doses of 250 mg/kg, 500 mg/kg and 1000 mg/kg; and a HFD accompanied by NFPUE treatment at a dose of 500 mg/kg. Adipose tissues show fewer lipid droplets (yellow arrow) and well-structured hepatocytes (black arrows) in the treated groups compared to the HFD group. Epididymal and mesenteric adipose tissue-stained images also reveal many small adipocytes in the FPUE-treated groups compared to the HFD group.

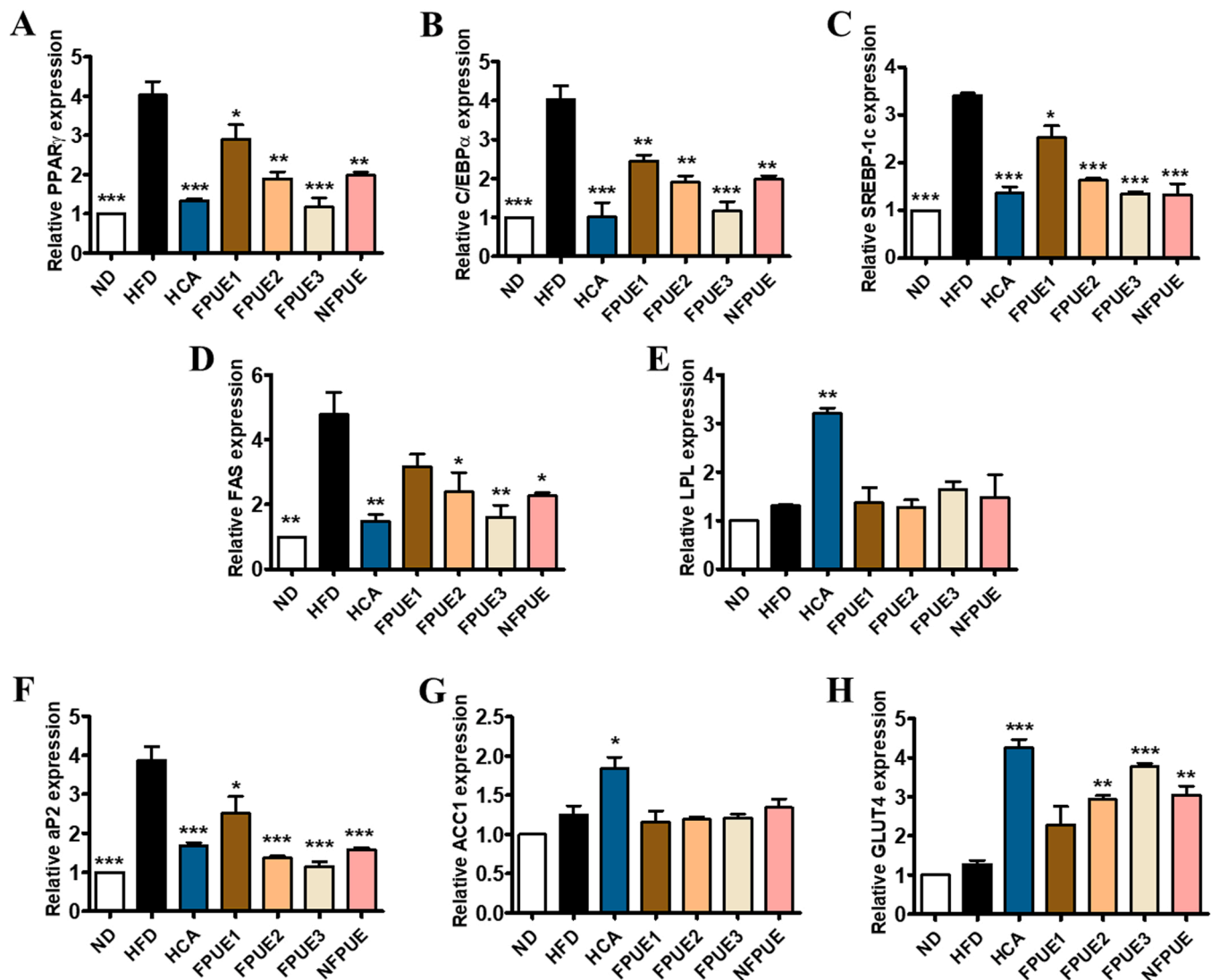


Fig. 8. *In vivo* effects of FPUE and NFPUE on adipogenic and lipogenic genes. Relative mRNA expression of adipogenic and lipogenic genes in HFD-fed rats following eight weeks of treatment with FPUE and NFPUE. The results are provided for: (A) PPAR-γ, peroxisome proliferator-activated receptor-γ; (B) C/EBP-α, CCAAT/enhancer binding protein-α; (C) SREBP-1c, sterol regulatory element-binding protein-1c; (D) FAS, fatty acid synthase; (E) LPL, lipoprotein lipase; (F) aP2, adipocyte fatty acid binding protein; (G) ACC1, acetyl-CoA carboxylase 1; and (H) GLUT4, glucose transporter 4. Values are expressed as mean ± SEM (n = 8). * ($p \leq 0.05$), ** ($p \leq 0.01$), and *** ($p \leq 0.001$), different from HFD group.

weight gain making the HFD-induced obesity model reliable for studying obesity. We observed a significant difference in body weight gain between HFD-fed and control chow-fed rats, although food intake was not significantly different between these two groups of animals. This finding indicates that the increased body weight gain in HFD-fed rats is not associated with the amount of food consumption. The feeding of FPUE (250–1000 mg/kg) reversely reduced the body weight gain remarkably in comparison to HFD only. This result suggests that the supplementation with FPUE can prevent the extra body weight gain, without inhibiting the general food consumption associated with weight gain.

Obesity alters the endocrine and metabolic functions of adipose tissue and leads to an increased release of hormones, fatty acids, and proinflammatory molecules contributing to obesity-associated complications [2]. Obesity can be described as a state of excessive growth of adipose tissue mass [47]. We thus measured the adipose tissue mass in this study and observed a significant increase in the weight of subcutaneous, mesenteric, perirenal, and epididymal adipose tissues among HFD-fed rats compared to control chow-fed rats, which was prevented by FPUE (250–1000 mg/kg of body weight). Subcutaneous and visceral

abdominal adipose tissues are major types of body fat. Visceral fat is composed of several adipose reserves, including epididymal, mesenteric, and perirenal ones. Abdominal obesity is correlated with an increased risk of cardiovascular diseases, and insulin resistance, whereas augmented subcutaneous fat is linked to a favorable plasma lipid profile. Mesenteric adipose tissue is considered to play a central role in the development of obesity-related diseases. Adiponectin and leptin are mainly available from adipocytes [47]. In the current study, the mean adipocyte mass for all the examined adipose tissues was noticeably reduced in FPUE-supplemented HFD-fed rats. The major cellular component of adipose tissue is the adipocyte, making the understanding of the mechanisms that regulate the number and size of adipocytes an important aim in obesity research [48]. Excessive energy is stored as triglyceride in adipocytes, while the expansion of adipocyte size (hypertrophy) reflects an elevation of intracellular lipid accumulation, dependent on the lipogenic activity of adipocytes. A critical aspect of the chronic situation of obesity is the increase of adipocyte cell number (hyperplasia), even though the initial event in obesity development is hypertrophy [49].

Leptin and ghrelin are two hormones recognized to have a major

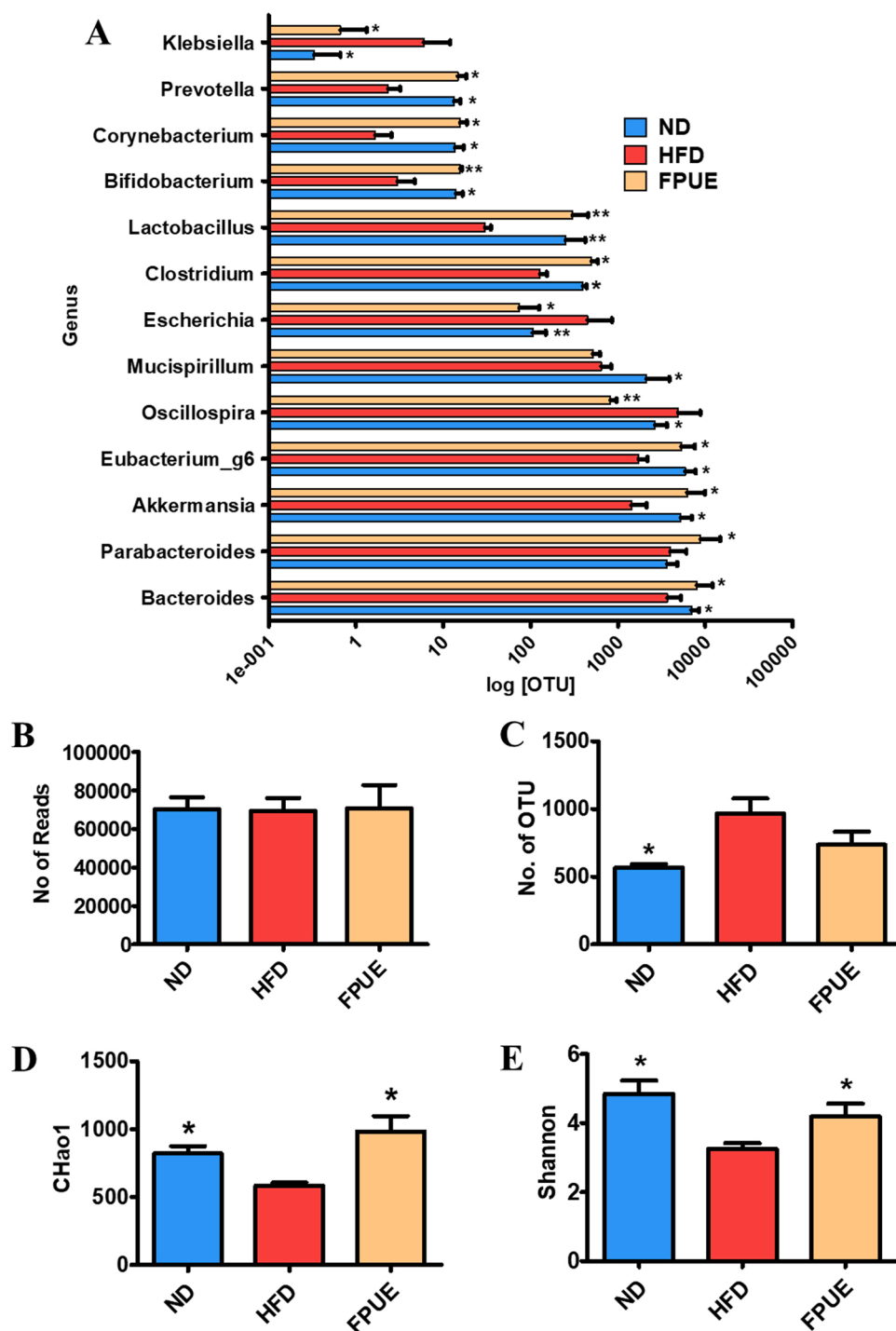


Fig. 9. Effect of FPUE on the intestinal microbiota major genera composition and phylogenetic diversity. FPUE treatment ameliorates the microbiota profiles in rats receiving a HFD. Bacterial genomic DNA was extracted from the feces samples, then 16S rDNA sequence analysis was performed. The panels show the: (A) relative composition of important intestinal bacteria at the genus level among the different groups; (B) the valid reads per group and alpha diversity analysis; (C) the Operational Taxonomic Units (OTUs); (D) the Chao1; and (E) the Shannon diversity index used for the evaluation. ND: normal diet; HFD: High-fat diet; FPUE: HFD + FPUE. Values are expressed as mean $\pm$ SEM ($n = 3$). * ($p \leq 0.05$), ** ($p \leq 0.01$), and *** ($p \leq 0.001$), different from HFD group.

influence on energy balance. The adipocyte hormone that controls body weight by modulating food intake and energy expenditure is leptin. The concentration of leptin is proportional to the percentage of body fat. Elevated levels of plasma leptin were identified in obese individuals compared to non-obese individuals [14]. In the current study, FPUE further suppressed the elevation of plasma leptin, which suggests that the observed reduction in body weight and fat mass are related to the reduction of leptin [50]. Ghrelin controls appetite and hunger by sending to the brain a signal indicating that the subject is hungry [14]. In this study, the levels of Ghrelin were significantly different for FPUE-supplemented versus non-supplemented HFD-fed rats, which indicates that FPUE influences the secretion and regulation of ghrelin.

In this study, apart from a reduction in body and adipose tissue weight, the supplementation of HFD with FPUE was observed to significantly attenuate the plasma levels of TC, FFA, AI, and LDL. The basic mechanism behind the reduced levels of TC, FFA, LDL, and AI might involve the breakdown of lipids by the extract components, thus normalizing lipid metabolism [51]. The reduction of TC, LDL, and FFA in plasma indicates a clearance of circulating lipids, which could protect cells from free radicals-induced damage [52]. Further, AI is considered a marker for various cardiovascular disorders, with its value being directly proportional to the risk of developing cardiovascular disease and vice versa [53]. Thus, high AI observed in the FPUE-treated groups appears beneficial, possessing lipid-lowering properties associated with a

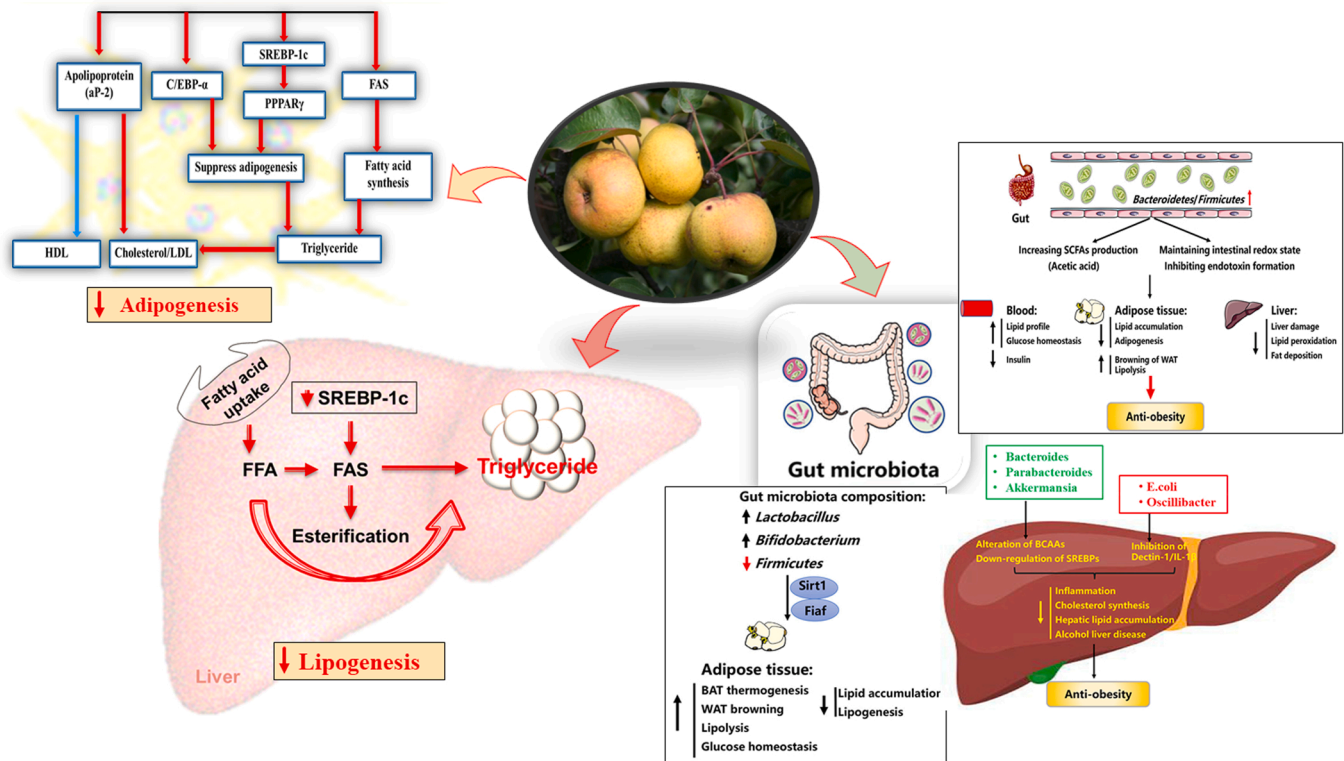


Fig. 10. Proposed mechanisms underlying the beneficial effects of FPUE against diet-induced obesity. The schematic pathway illustrated indicates that FPUE exerts its therapeutic effects against obesity either by reducing lipogenesis and adipogenesis via an upregulation (black) and downregulation (red arrows) of several transcriptional factors and/or through modulation of the intestinal microbiota via increased (black arrows) and decreased (red arrows) microbial proportions.

cardio-protective potential. In addition, the histopathological investigation showed a protective effect of FPUE from inflammation and steatosis which further consolidate the biochemical assay findings.

Our results show that FPUE significantly inhibits preadipocyte differentiation by attenuating the MDI-induced lipid accumulation. Master transcriptional regulators of adipogenesis, including PPARγ, C/EBPα, and SREBP-1c, are necessary to modulate gene expression involved in adipogenesis and differentiation at various stages of these processes [11, 54]. Interestingly, the feeding of HFD *versus* control chow to rats significantly increased their mRNA expression of PPARγ, C/EBPα, and SREBP-1c genes. The mRNA expression of these genes in the liver tissues of FPUE-supplemented HFD-fed rats was normalized to control levels, thereby reducing the expression of their target genes, including FAS, and aP2, while increasing the expression of GLUT4. In the past few decades, the role of PPARγ and SERBP-1c in the regulation of obesity and adipocyte differentiation was highlighted and hence, PPARγ agonists and antagonists of synthetic or herbal origin have gained wide commercial popularity as therapeutic agents [55]. Moreover, PPARγ regulates the expression of adipocyte genes such as adipocyte-fatty acid binding protein (A-FABP) [56], while the SREBP-1c controls the expression of the lipogenic genes FAS and ACC-1, in addition to activating insulin-mediated transcriptional factors [50]. In the present study, a significant downregulation of PPARγ, C/EBPα, and SREBP-1c mRNA expression was measured following FPUE supplementation, in agreement with previous findings [14,55].

The FAS is an enzyme related to fatty acid metabolism, while LPL is linked to the cellular intake of fat [57]. We observed tendencies for reduced FAS in the FPUE-treated group compared to the HFD group; however, we did not find a significant reduction in the mRNA expression of ACC-1, which is controlled by SREBP-1c, and LPL. These results show that the anti-obesity effect of FPUE may result from an attenuated fatty acid metabolism and reduced lipogenesis genes, rather than a decreased uptake of lipids in the liver. Additionally, the aP2 as a carrier protein and

facilitator reduced plasma lipid levels (LDL and total cholesterol) by increasing HDL, thus providing an alternative mechanism to cholesterol clearance [58]. The PPARγ which regulates GLUT4 is one of the main insulin-responsive glucose transporters, more related to insulin resistance and diabetes (type 2) than obesity [37]. In the present study, FPUE supplementation significantly enhanced GLUT4 mRNA expression and normalized elevated levels of insulin in HFD-fed rats. Hyperinsulinemia accompanied by obesity is considered a biomarker of insulin resistance [37]. Thus, the present findings show that FPUE can prevent insulin resistance during obesity.

Many previous studies indicate that apart from metabolic pathway regulation, obesity is closely related to intestinal microbiota. Lean phenotypes and obese individuals who lose weight have higher amounts of *Bacteroidetes* in their gut microbiota [59]. A number of studies have reported an association between changes in the relative abundance of *Bacteroidetes* to *Firmicutes* ratio and obesity [60,61]. Consistent with these findings, our results demonstrate that the *Bacteroidetes* phylum was significantly attenuated, while the *Firmicutes* phylum was significantly enhanced in the HFD-induced obese rat model. Accordingly, FPUE administration prevented diet-induced obesity and reversed the relative abundance of gut microbiota in HFD-fed obese rats. Lecomte et al. revealed that the increased abundance of *Firmicutes* in the intestinal microbiota of obese patients may promote more efficient absorption of calories and reduced weight gain through an enhanced capacity to harvest energy from the diet [62]. Furthermore, a higher proportion of *Firmicutes* instead of *Bacteroidetes* was observed in diet-induced or genetically obese mice [63]. This increased proportion is linked to increased secretion of enzymes producing short-chain fatty acids (SCFA) from the catabolism of indigestible polysaccharides in the diet [64,65]. These SCFAs increase the browning of white adipose tissue (WAT) and glucose homeostasis in blood, while decreasing lipid peroxidation, insulin, and lipid accumulation in the liver, blood, and adipose tissue, respectively [63]. *Lactobacillus* and *Bifidobacterium* are frequently used

as probiotics, and previous studies have shown an increased proportion of *Lactobacillus* and *Bifidobacterium* in obese individuals [59,66]. In another report, Lecomte et al. have shown that the abundance of these two species was negatively correlated to body weight and fat mass [62]. Our study findings are consistent with those from the Lecomte report. In addition, FPUE treatment had a positive impact on another well-known intestinal bacteria, the genus *Prevotella*, suggesting an additional mechanism by which FPUE might regulate relative metabolism. Moreover, this enhanced *Lactobacillus* and *Bifidobacterium* proportion following treatment with FPUE might contribute to an increased WAT browning, named brown adipose tissue thermogenesis, as well as glucose homeostasis and lipolysis. It was reported that HFD-fed rats have an increased proportion of *Oscillibacter* spp. showing a negative connection with the expression of intestinal tight junction proteins [67], in line with our results. In the current study, FPUE treatment significantly decreased the *Oscillospira* genus proportion in HFD-fed rats. Previous findings revealed that *Clostridium* cluster bacteria modulate the host fatty acid metabolism, as well as induce Treg cell activity and attenuate colitis [68]. Moreover, the *Eubacterium* spp. induced with the prebiotic oligosaccharides produced beneficial effects on hosts [59]. Our findings show that these two genera increased significantly in FPUE-treated rats. Previous research reported that *Eubacterium* and *Clostridium* could produce butyrate, which has an anti-obesity effect [59]. These results indicate that the effects of FPUE treatment on reducing body weight gain and fat accumulation might result from increasing the population of these beneficial species. Furthermore, obesity reduced the microflora diversity and overall balance of intestinal microflora which is important for homeostasis [69]. Our findings provide similar results, by indicating that HFD consumption in rats reduces the intestinal microflora diversity while FPUE treatment normalized this reduction. Furthermore, our FPUE treatment shifted the gut microbiota relative abundance in HFD-induced obese rats, successfully reduced their weight, and regulated their metabolism.

5. Conclusion

In conclusion, our results suggest that FPUE has a direct effect on the endocrine system and fat accumulation, notably by influencing the intestinal microflora. The outcomes of this extract led to decreased lipogenesis, body weight gain, fat mass, adipocyte size, and hepatic abnormalities. While multiple mechanisms may play a crucial role in reducing the HFD-induced obesity we observed, the relationship among these mechanisms remains to be further investigated. This study is the first scientific report providing strong evidence to support the relevance of using fermented *Pyrus ussuriensis* Maxim as a superfood with anti-obesity properties. Together, our results indicate that the fermented PUE represents a promising candidate for the prevention and treatment of obesity.

CRedit authorship contribution statement

Naila Bobby, Seung-Chun Park: Conceptualization. **Naila Bobby, Muhammad A. Abbas, Eon-Bee Lee, Zi-Eum Im:** Methodology. **Naila Bobby, Muhammad A. Abbas:** Investigation. **Naila Bobby:** Software. **Naila Bobby:** Data curation. **Zi-Eum Im:** Resources. **Naila Bobby:** Formal analysis. **Naila Bobby:** Writing – original draft. **Naila Bobby, Seung-Chun Park:** Writing – review & editing. **Seung-Jin Lee, Seung-Chun Park:** Supervision, Project administration, Funding acquisition.

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dissertation conducted at Kyungpook National University.

Declaration of Competing Interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence that work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2022.113629.

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